

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458195.6616 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.016
Transit depth (Rp/Rs)^2	1.8 +/- 0.42 [%]
Orbital Inclination (inc)	84.51 +/- 1.19
Airmass coefficient 1 (a1)	0.92 +/- 0.013
Airmass coefficient 2 (a2)	0.062 +/- 0.054
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#2 - [241, 269]
Optimal Aperture	2.85
Optimal Annulus	10.21
Transit Duration (day)	0.0807 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.134 ± 0.016 vs 0.1368 (deviation 0.16σ)
Duration fit vs published	0.0807 d vs 0.0758 d (deviation 0.79σ)
Mid-time O–C	4.7 min
Depth SNR	7.6
Residual scatter	1.42 %

Light curve

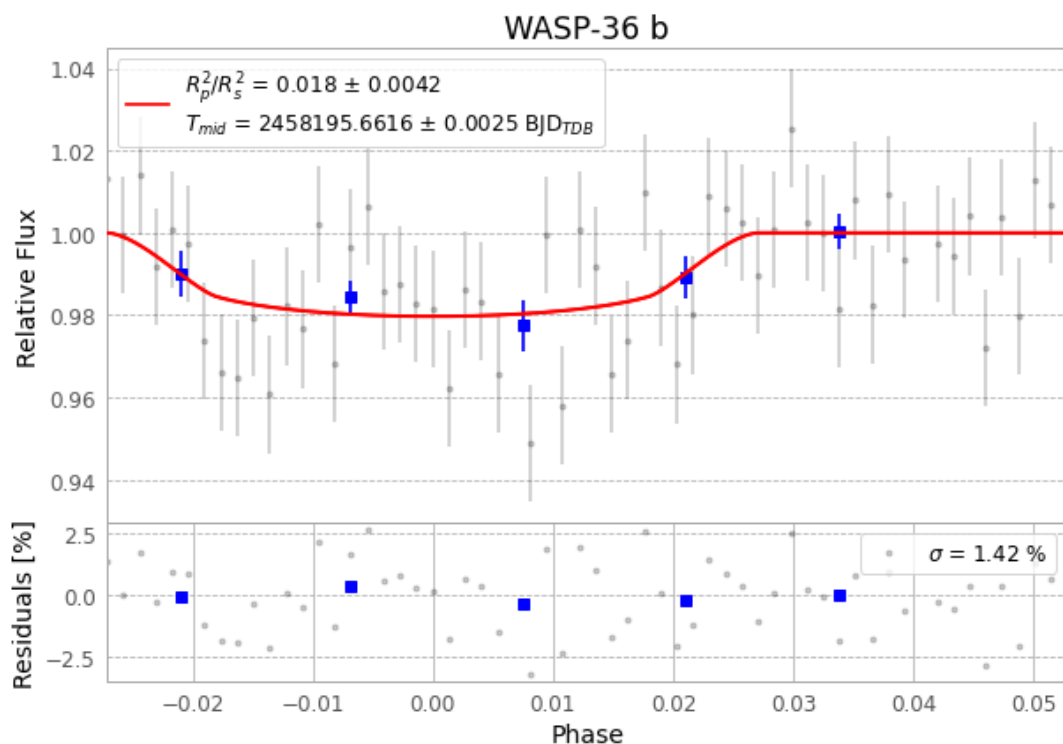


Figure 1 — FinalLightCurve_WASP-36 b_2018-03-17.png (SHA-256 99dae20ec0fcfb8b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2240aa3d26d286e9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

60 FITS frames (39 MB), WASP-36180318024512.FITS → WASP-36180318054212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180318.log (1637fe24eba7...), FinalLightCurve_WASP-36 b_2018-03-17.png (99dae20ec0fc...), FinalLightCurve_WASP-36 b_2018-03-17.csv (ebcdd0cb28f2...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 22:55Z.